

Positioning the Infiniti G20 Case

With Technical Appendix on PCA and Supplementary Variables in R

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Introducing the G20

In April 1990, Nissan's Infiniti division planned to introduce the G20 in the U.S., adding a third model to the existing Infiniti line. The G20 was already available in Europe and Japan under the name Primera. The car, equipped with a four-cylinder engine developing 140 horsepower, would be Infiniti's entry-level luxury car. Initial market response to the G20 in the U.S. was disappointing, and management wondered how it might retarget or reposition the car to improve its market performance.¹

Background

In 1989, three years after Honda first introduced its Acura line, Toyota and Nissan attacked the U.S. luxury car market, a segment previously dominated by American and German manufacturers.

In November 1989, Nissan launched its new luxury Infiniti division with the \$40,000 Q45 as its lead car and the \$20,000 M30. However Nissan was somewhat late: In August 1989, three months before Nissan shipped its first Infiniti, Toyota had introduced Lexus, its luxury brand with a two-car line comprising the \$40,000 LS400 and the entry-level LS250.

As the figures for January to September 1990 showed, Lexus outsold Infiniti by 50,000 to 15,000. The reasons for Infiniti's slow start were threefold.

- First the Infiniti Q45 came to the market after the Lexus LS400 had established a good market position.
- Second Lexus had two very good cars, whereas Infiniti's M30 coupe received poor evaluations from the automobile press and from customers.

¹Data of the case in the G20.csv file.

- Finally the eccentric Infiniti advertising campaign that showed scenes of nature, but not the car itself, shared some of the blame. (“Infiniti may not be doing so well, but, hey, at least sales of rocks and trees are skyrocketing,” commented comedian Jay Leno.)

Research data

Exhibits 1–4 summarize some of the data that Infiniti had in early 1990. Data in Exhibits 1 and 2 are based on a survey of customers from its target segments, described as people between 25 and 35 with annual household incomes between \$50,000 and \$100,000 (when the survey was administered, the Lexus LS250 was not yet well known to the respondents to be included in the study). The three sub-segments in Exhibit 1 (denoted S1, S2, and S3) are based on information provided by Infiniti managers. Exhibit 3 is derived from sales brochures describing the characteristics of each car. Exhibit 4 summarizes demographic and psychographic information about the three sub-segments and was compiled from databases supplied by Claritas, Inc.

In this TeX version, the numerical contents of Exhibits 1 and 2 (perceptions and preferences) are provided in the external file `g20.csv`, described in the technical appendix.

Exercises

1. Describe the two (or, if applicable, three) dimensions underlying the perceptual maps that you generated. Based on these maps, how do people in this market perceive the Infiniti G20 compared with its competitors?
2. What are the average car ratings for each of the 3 segments? Represent these average preferences on the perceptual map using the option to add additional variables in the representation.
3. Infiniti promoted the G20 as a Japanese car (basic version \$17,500) with a German feel, basically a car that was like the BMW 318i (\$20,000), but lower priced. Is this a credible claim, given the perceptions and preferences of the respondents?
4. Which attributes are most important in influencing preference for these cars in the three segments (S1, S2, and S3) shown on these maps? To which segment(s) would you market the Infiniti G20? How would you reposition the Infiniti G20 to best suit the chosen segment(s)? Briefly describe the marketing program you would use to target the chosen segment(s).

5. What ongoing research program would you recommend to Infiniti to improve its evaluation of its segmentation of the market and positioning of its G20?
6. Summarize the advantages and limitations of the software provided for this application.

	<i>G20</i>	<i>Ford T-bird</i>	<i>Audi 90</i>	<i>Toyota Supra</i>	<i>Eagle Talon</i>	<i>Honda Prelude</i>	<i>Saab 900</i>	<i>Pontiac Firebird</i>	<i>BMW 318i</i>	<i>Mercury Capri</i>
Base Price \$	17,500	15,783	20,200	23,280	16,437	14,945	18,295	12,690	19,900	13,500
Length	175	198.7	176	181.9	170.5	175.6	184.5	192.0	170.3	166.1
Width	66.7	72.7	67.6	68.7	66.7	67.3	66.5	72.4	64.8	64.6
Height	54.9	52.7	54.3	51.2	51.4	29.2	56.1	49.8	53.5	50.2
Weight (lbs.)	2,535	3,600	3,170	3,535	3,100	2,740	2,825	3,485	2,600	2,487
Fuel Economy (Mpg)										
City	24	17	18	17	20	23	20	16	22	23
Highway	32	24	24	22	25	27	26	24	27	28
Horsepower, SAE.net	140@	210@	164@	232@	195@	135@	140@	240@	134@	132@
(Bhp)	6,400	4,000	6,000	5,600	6,000	6,200	6,000	4,400	6,000	6,000
Warranty, Years/Miles	4/60,000	1/12,000	3/50,000	3/36,000	1/12,000	3/36,000	3/36,000	3/50,000	3/36,000	1/12,000

EXHIBIT 3: Some physical characteristics of the cars.

<i>Segment Characteristic</i>	<i>Segment I (Upwardly Mobile Families)</i>	<i>Segment II (Western Yuppie, Single)</i>	<i>Segment III (American Dreamers)</i>
Segment Size	(45%)	(25%)	(30%)
Education	College Grads or Some College	College Grads	College Grads or Some College
Predominant Employment	White-Collar	Professionals	White-Collar
Age Group	25-35	25-35	25-35
Predominant Ethnic Background	White	White	Mix (Asian, White)
Average Household Income	\$68,000	\$81,000	\$59,000
Persons per Household	3.8	1.42	2.4
Percent Married	75%	32%	55%
Watch Late Night TV	9%	27%	17%
Watch Daytime TV	45%	3%	5%
Read Computer Magazines	6%	39%	10%
Read Business Magazines	23%	58%	27%
Read Entertainment Magazines	14%	3%	30%
Read Infant and Parenting Magazines	17%	1%	2%
Rent Movies	85%	43%	38%
Possess an American Express Card	45%	48%	75%
Own Investment Funds	18%	24%	47%
Go Fishing	30%	2%	3%
Sail, Scuba Dive or Ski	2%	49%	20%

EXHIBIT 4: Data about the sub-segments.

Technical Appendix: Supplementary Variables in PCA

In this case, principal component analysis (PCA) is performed on the 15 perceptual attributes. However, managers may also want to visualise additional information—such as the average preferences of the three segments (S1, S2, S3) or other external measures—on the perceptual map. These extra variables should not influence the axes of the PCA, because the perceptual structure must be determined only by the original attributes.

Supplementary variables solve this problem: they are projected onto the PCA space *a posteriori*, so they help interpretation without modifying the dimensions.

Below are the commands used to perform this projection in R.

Required R Packages

To reproduce the PCA and generate biplots with supplementary variables, two packages must be installed and loaded:

- **FactoMineR** — provides the `PCA()` function used to compute the analysis with active and supplementary variables.
- **factoextra** — provides convenient visualisation tools, including `fviz_pca_biplot()` which displays objects, active variables, and supplementary variables on the same map.

The necessary commands are:

```
# Install packages (only once)
install.packages("FactoMineR")
install.packages("factoextra")

# Load packages (each session)
library(FactoMineR)
library(factoextra)
```

Both packages must be loaded before running the PCA or producing the biplot.

PCA with supplementary quantitative variables

```
# PCA with 15 active variables + 4 supplementary variables (segment means)
res_pca <- PCA(
  data_acp[c(1:15, 91:94)], # 1:15 = perceptual attributes
```

```
# 91:94 = segment means or other supplementary vars
scale.unit = TRUE,
quanti.sup = c(16:19),      # positions of supplementary variables
graph = TRUE
)
```

Explanation:

- Only the first 15 variables (the perceptual attributes) determine the PCA axes.
- Columns 91–94 are included in the data frame but declared in `quanti.sup`: they are *projected* onto the PCA, not used to compute it.
- This allows segment preferences (or any external measures) to be displayed on the map without changing the perceptual structure.

Displaying objects, variables, and supplementary variables

```
# Biplot showing:
# - cars (objects)
# - perceptual attributes (active variables)
# - supplementary variables (segment means)
fviz_pca_biplot(res_pca)
```

This command produces a joint map where:

- cars appear as points,
- perceptual attributes appear as vectors,
- supplementary variables (e.g., S1, S2, S3) appear as additional arrows, helping to interpret which parts of the map correspond to each segment's preferences.

This is all that is required to represent supplementary variables in the G20 PCA analysis.